

X-haul solutions for 5G/6G networks: Overview of requirements and technologies

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ABSTRACT

This paper is an extended version of the FOAN 2024 invited paper titled Overview of Available Fiber Optic Backhaul Solutions for 5G/6G Networks. With cloud-RAN, backhaul now includes fronthaul and midhaul, forming x-haul that links remote radio heads to baseband units and interconnects nodes. Evolving networks demand higher bandwidth, lower latency, and improved reliability. Ultra-dense 5G/6G deployments require advanced solutions like TWDM-PON, supporting 25 Gbps for 5G cells and over 100 Gbps for 6G mmWave clusters, with latency under 100 μ s. The main contribution of this paper is a comprehensive overview of x-haul requirements and available technologies.

1. Introduction

The fifth generation of mobile networks (5G) aims to provide “connectivity anywhere, at any time,” driving the development of ultra-dense heterogeneous networks (HetNets) with multi-layered topologies and various radio access technologies (multi-RAT) integration. While the radio access network (RAN) was traditionally viewed as the primary bottleneck from capacity point of view, the backhaul has emerged as equally critical, requiring alignment with RAN demands and specific use cases.

The backhaul is a key part of RAN architecture, connecting network elements like Base Station Controller (BSC) in 2G, Radio Network Controller (RNC) in 3G, Evolved Node B (eNB) in 4G, and Next-Generation Node B (gNB) in 5G to the core network via technologies such as copper, fiber optics, microwave, and mmWave. With the introduction of cloud-RAN (C-RAN) architecture, backhaul is redefined into three layers: fronthaul, midhaul, and backhaul, collectively referred to as x-haul in this paper.

The fronthaul connects the remote radio head (RRH) to the baseband unit (BBU) using protocols like Common Public Radio Interface (CPRI) or its enhanced version eCPRI, supporting user plane, synchronization, and operations, administration, and management (OAM) functionalities. An alternative solution to the CPRI protocol is the usage of Open Base Station Architecture Initiative (OBSAI) solution. The midhaul links multiple eNBs or gNBs via the X2 interface, bridging fronthaul and backhaul aggregation points. The backhaul, through the S1 interface, connects these aggregation points to the core network.

The comparison of C-RAN and distributed RAN (D-RAN) architectures is shown in Fig. 1. They serve different use cases and are deployed

based on specific needs. Fog radio access network (F-RAN) is another architecture that combines the strengths of C-RAN and fog computing, addressing mobile traffic growth and enhancing Quality of Service (QoS) [1]. It leverages caching and artificial intelligence (AI) at the network edge, enabling real-time signal processing, flexible networking, and optimized capacity and latency in the fronthaul and BBU pool [1].

5G use cases fall into three categories: enhanced mobile broadband (eMBB), offering high bandwidth and high quality internet access expected to range from 100 Mbps up to 1 Gbps user data rates; massive machine-type communication (mMTC), enabling low-rate, predominantly uplink non-human communication; and ultra reliable low latency communication (URLLC), focusing on high reliability and low latency [2]. Looking ahead, the sixth generation of mobile networks (6G) aims to achieve latency below 1 ms, data rates up to 10 Tbps, high three-dimensional coverage, massive connectivity, extremely high reliability (99,9999%) and improved energy and spectral efficiency [1]. Both 5G and 6G are high-density networks, which is a critical consideration for x-haul planning.

6G is expected to introduce new service types beyond 5G use cases. Computation-oriented communications (CoC) will flexibly adapt data rate, latency, and reliability based on resource availability to ensure computational accuracy [3]. Contextually agile eMBB communications (CAeC) will enhance eMBB by adapting to network, physical, and social contexts. It accounts for various contexts, including network congestion and topology, physical factors like location and mobility, and social aspects such as neighborhood and sentiments [3]. Event-defined URLLC (EDURLLC) will support URLLC in dynamic emergency scenarios, addressing varying device densities, traffic patterns, and resource availability [3].

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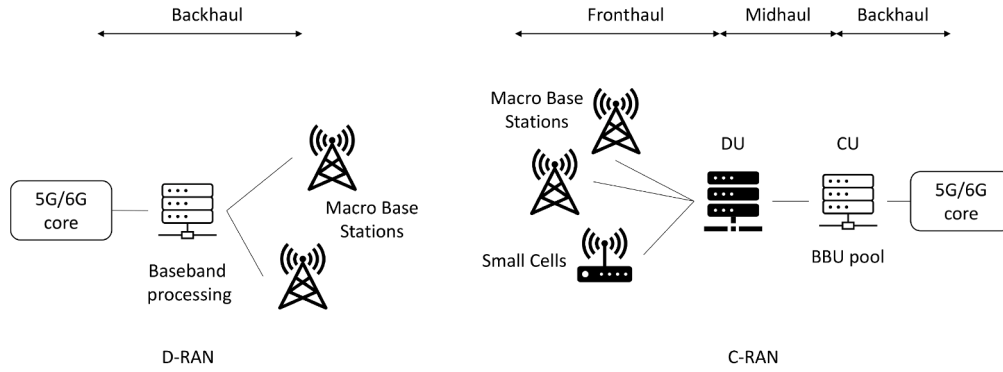


Fig. 1. The comparison of D-RAN and C-RAN architectures.

When dimensioning fronthaul links, it is essential to consider all key parameters such as channel bandwidth, number of antennas (Multiple Input Multiple Output - MIMO), aggregated user traffic, and the selected level of functional centralization (functional split). The choice of split has a significant impact on both required capacity and latency. The underlying x-haul technology must meet the diversity of requirements of the heterogeneous network with the ability to dynamically adjust and adapt to the changes in the network in a flexible, efficient, and timely manner [4].

This paper is an extended and thoroughly revised version of a paper titled Overview of Available Fiber Optic Backhaul Solutions for 5G/6G Networks, that was initially presented at the 2024 FOAN (Fiber Optics in Access Networks) conference [5]. The first conference submission focused exclusively on fiber optic backhaul solutions and it provided a foundation for the current literature review but received critical feedback, particularly regarding the breadth of quantitative requirements and the need for more detailed coverage of wireless and emerging optical x-haul solutions. Insights and recommendations from conference reviewers and audience discussions played a pivotal role in improving the scope, structure, and technical depth of the manuscript, ensuring that the present work better addresses open research questions and enhances replicability.

To ensure comprehensive coverage of the state-of-the-art in x-haul technologies for 5G/6G networks, a systematic literature review was performed. Searches were conducted in major scientific databases including IEEE Xplore, Scopus, Web of Science, ScienceDirect, and Google Scholar, primarily for the period 2015 - 2025. The following keywords and their combinations were used: “x-haul”, “fronthaul”, “midhaul”, “backhaul”, “5G/6G transport”, “optical passive networks”, “PON 5G”, “dynamic bandwidth allocation PON”, “network slicing x-haul”, “SDN optical x-haul”. In addition, relevant international standards (ITU-T and 3GPP) and white papers were included [2,6–12]. Only peer-reviewed papers, major standards, or impactful industry reports were considered, and works were screened for technical rigor, quantitative data, and practical relevance.

The rest of the paper is organized as follows. Section 2 provides an overview of the main x-haul requirements that meet 5G/6G RAN standards, while available wired and wireless x-haul solutions are listed and compared in Section 3. The fiber optic x-haul solution is in the focus of Section 4, giving an overview of the deployed topologies, enabling fiber optic technologies and, finally, passive optical networks (PON) solutions. This last topic is of particular interest to this section before the final conclusions are given.

2. X-haul requirements

Each generational change of the mobile access network standards has had direct consequences on x-haul design. The x-haul always had to

follow traffic requirements originating from the mobile access network and/or core network segment. The evolution from the first to the fifth generation of the technological standard for cellular networks has been accompanied by increasingly strict requirements for the x-haul segment of the architecture. The initial driver was speed (bandwidth), but it was soon supplemented by additional requirements such as latency, bit error ratio, massive device connectivity, reliability, high quality of experience (QoE) and, finally, spectral and energy efficiency. 5G networks are characterized by ultra dense cellular networks providing seamless coverage supported by substantial increase in bandwidth, including MIMO method for multiplying the capacity of a radio link. The introduction of massive MIMO (mMIMO) and coordinated multipoint (CoMP) technologies requires very low x-haul latency followed by high bandwidth to follow small cell growth in ultra dense networks [4].

The radio signal processing functions occur at the RRH site. The quality of the radio frequency (RF) signal amplified at the RRH depends heavily on the attenuation of the coaxial cable from RRH reaching the antenna. Consequently, the main requirement is to assure minimum degradation of the RF signal and to assure a reach that is long tens of meters of propagation. The utilization of DAC & ADC (Digital to Analog Converter and vice versa) and digital transmission over fiber with regular pluggable optoelectronic transceivers satisfies these requirements [13].

Digitized time domain radio signals that originate from the BBU, are linked to the RRH via fronthaul through constant and symmetrical high bit rate serial digital interfaces [13]. To calculate necessary fronthaul link capacity it is required to know desired bandwidth, applied MIMO scheme, the number of sectors of the observed mobile network infrastructure point as also its maximum peak bit rate. Quantization and coding operations needed for the translation of the radio signals into on-off keying (OOK) sequences used in the optical fronthaul link cause poor spectral efficiency [13]. As the clock of this OOK signal serves as a reference for mobile RF generation inside the RRH, the requirements for the fronthaul reach are satisfied thanks to the optical transceivers that have a range of tens of kilometers at the required bitrates [13]. This also ensures that fronthaul latency requirements are met.

2.1. Functional splits

Functional split options in wireless communication systems enhance flexibility and efficiency by dividing the physical (PHY) layer between the Radio Unit (RU) and Distributed Unit (DU). To summarize the key functional split options and their corresponding x-haul requirements [14]:

- Option 7.1: offers high performance but requires significant bandwidth and low latency, with data rates scaling with MIMO layers.

- Option 7.2: Similar to 7.1, but combines signals from multiple antenna ports, reducing bandwidth requirements while maintaining performance.
- Option 7.3: Reduces fronthaul bit rates significantly by moving functions like modulation and demodulation closer to the RU. While it lowers capacity demands (e.g. considering a cell bandwidth of 10 MHz or 20 MHz, 2x2 MIMO, 4 antenna ports, and 16-QAM modulation 134.4 Mbps for Option 7.3 is required in comparison with 1075.2 Mbps for Option 7.2 and 4300.8 Mbps for Option 7.1), it increases RU complexity.

Generally speaking, various functional split options come along with their corresponding requirements for bandwidth, latency, and possible implementation technologies. Split options have latency requirements ranging from 10 ms (Option 1) to 200 μ s (for advanced options like 4–8), while bandwidth demands increase significantly with higher options, from 1 Mbps (Option 1) to as high as 885 Mbps (Option 8) [14]. Point-to-point links (P2P), PON, and Free Space Optics (FSO) are available x-haul solutions for split options requiring lower latency and greater flexibility (Options 4 and 5), however, advanced options like 7.1 and 8 rely on P2P x-haul due to high bandwidth requirements [14]. Advanced split options deliver better performance but require more sophisticated infrastructure with higher capacity and lower latency [14]. Efficient x-haul is not just critical for meeting high data rate and low latency demands in 5G and beyond, but it is also vital for supporting macro/small cells, distributed antenna systems, and RAN virtualization.

2.2. X-haul challenges in terms of mobile network KPIs

The major x-haul challenges regarding 5G networks can be categorized into eight different groups [15,16]:

- Network capacity: 5G network interconnects ultra dense network comprised of numerous smart devices to other mobile connections. 5G network standards require increased capacity of x-haul links for each cell site. Fronthaul link data rate in particular is proportional to the number of antennas per sector, sampling rate (sample/s/carrier) of the used fronthaul protocol (i.e. CPRI, OBSAI, etc.), sample width (bits/sample) and fronthaul coding factor [16]. The fronthaul network must maintain signal integrity to avoid degrading radio performance, which is primarily measured by the Error Vector Magnitude (EVM) at the RU output [16]. As an example, if these parameters are considered for a radio access network with the functional split option 7 operating in the mmWave frequency range at 3.5 GHz with 100 MHz bandwidth and a 32×32 MIMO configuration, assuming a sample width of 15 bits/sample, a CPRI control word factor of 16/15, a coding factor of 66/64 (for 64B/66B coding), and a digitization sampling rate of 153.6 MHz, the resulting required CPRI rate per sector is approximately 162.2016 Gbit/s.
- Network configuration: network densification assumes increased density of small base stations (SBSs), supporting an increased number of users. In order to satisfy this requirement, it is of vital importance to provide high capacity x-haul connectivity to SBSs. Macro cells typically have three to six sectors, with each sector supporting multiple Radio Access Technologies (RATs) across different frequency bands that can result in up to 15 Remote Units (RUs) per cell site. To minimize the number of fibers required to connect these RUs to the Central Office (CO), multiplexing techniques (time or wavelength) are necessary [16]. In contrast, micro/small cells typically use omnidirectional antennas, with only one RU per RAT and frequency band, simplifying their configuration compared to macro cells [16].
- Frequency: although the utilization of higher frequencies enables higher data rates required for typical eMBB use case scenarios, it comes at the expense of the cellular coverage. This brings us back to the second point, namely to the densification aspect of the technology. However, the densely deployed networks limit the frequency

reuse capability. Therefore, it is important to effectively utilize wireless x-haul spectrum.

- Availability: x-haul plays a crucial role in enabling high levels of availability in mobile access networks. This particularly concerns wireless x-hauls, especially in millimeter wave (mmWave) and microwave frequency ranges, as they are highly affected by adverse weather conditions and multipath propagation.
- Ultra-low latency: to decrease overall latency it is important to satisfy reliability and ultra dense network requirements. In the x-haul segment the balance needs to be made between the best possible solution from the technological point of view based on fiber optics, which offer huge capacity accompanied by very high levels of reliability and more cost-effective solutions based on wireless x-haul solutions. Fronthaul latency is not explicitly defined by RAN standards because it falls within the implementation-dependent part of the network (i.e. eNB or gNB) [16]. However, timing constraints are indirectly derived from RAN timing requirements, especially those related to Timing Advance (T_{ADV}), which is a mechanism that allows the base station (i.e. eNB or gNB) to adjust when the user equipment (UE) should start transmitting its data so that it arrives at the correct time, synchronized with other devices in the network [16]. The maximum fronthaul round-trip time (RTT) should remain well below the maximum T_{ADV} , with practical targets such as 500 μ s or even 200 μ s in legacy systems, accounting for fiber and equipment delays [16]. Accuracy of fronthaul RTT must be better than ± 130 ns, matching UE timing adjustment precision. Long-term latency variation (wander) due to factors like temperature must be limited, and standards such as IEEE 802.3 are proposed as benchmarks to manage such time drift [16].
- Network energy consumption: includes a reduction in carbon footprint, energy consumption and extended battery life. The passive character of fiber-to-the-home solution in x-haul links certainly has a potential to satisfy this kind of requirement.
- Deployment cost: a cost-effective x-haul solution has a substantial share in overall deployment cost of the 5G cellular networks. As previously mentioned, wireless x-haul networks prove to be more cost-efficient solution, but at the expense of several other requirements.
- Coverage: in ultra dense networks, particularly in urban areas, the deployment of fiber optic x-haul solutions can be thought as a promising solution since the convergence with fixed access network is possible. This is especially true for integrated service providers offering fixed and mobile services, where the convergence with the Fiber-to-the-Home (FTTH) network is even more cost-efficient solution. The costs of providing x-haul links can be shared with the costs of deploying the underlying fiber optic infrastructure that generates financial revenues from commercial services. Under different circumstances, i.e., in less populated areas, where there are no commercial reasons to deploy fiber optic network, various wireless solutions exist, but in that case, it is important to avoid interference and congestion in the x-haul networks.

Table 1 summarizes the fronthaul, midhaul, and backhaul requirements for two major KPIs, bandwidth and latency, across small cell, macro cell, and ultra-dense deployment scenarios. As previously stated, these values depend on the functional split, the number of antenna ports, and the aggregated frequency bandwidth. Typical values are provided based on relevant research, while projections for 6G are approximate, as the standards for 6G have not yet been finalized [1,14,17–19]. The deployment scenarios are based on the 3GPP Study on scenarios and requirements for next generation access technologies [12]. Small cell scenario satisfies the 3GPP indoor hotspot deployment scenario that is focused on high capacity (mid band and/or mmWave carrier frequency range) and small coverage per site as also high user throughput or user density in buildings (up to 200 MHz and 1 GHz aggregated bandwidth at mid band and mmWave frequency range, respectively). The key characteristics of this deployment scenario are high capacity, high user

Table 1

Bandwidth and latency requirements for fronthaul, midhaul, and backhaul across deployment scenarios in 5G and 6G networks [1,14,17–19].

X-haul Segment	Small cell	Macro cell	Ultra dense
Fronthaul	10–20 Gb/s and $\leq 100 \mu\text{s}$; 40–80 Gb/s and $\leq 50 \mu\text{s}$ (6G)	20–40 Gb/s and $\leq 100 \mu\text{s}$; 80–120 Gb/s and $\leq 50 \mu\text{s}$ (6G)	30–50 Gb/s and $\leq 100 \mu\text{s}$; 120–250 Gb/s and $\leq 50 \mu\text{s}$ (6G)
Midhaul	5–10 Gb/s and 1–3 ms; 20–30 Gb/s and $\leq 1 \text{ ms}$ (6G)	10–20 Gb/s and 1–3 ms; 40–80 Gb/s and $\leq 1 \text{ ms}$ (6G)	20–40 Gb/s and 1–3 ms; 80–150 Gb/s and $\leq 1 \text{ ms}$ (6G)
Backhaul	1–5 Gb/s and $\leq 10 \text{ ms}$; 10–20 Gb/s and $\leq 5 \text{ ms}$ (6G)	10–20 Gb/s and $\leq 10 \text{ ms}$; 100–400 Gb/s and $\leq 5 \text{ ms}$ (6G)	20–50 Gb/s and $\leq 10 \text{ ms}$; $\geq 400 \text{ Gb/s}$ and $\leq 5 \text{ ms}$ (6G)

density, typical inter-site distance (ISD) of 20 m and consistent user experience indoor [12]. Macro cell scenario is based on 3GPP urban macro deployment scenario that is focused on large cells and continuous coverage (mid band and mmWave carrier frequency range, enabling up to 1 GHz aggregated bandwidth). The key characteristics of this scenario are continuous and ubiquitous coverage in urban areas. This scenario will be interference-limited, using macro transceivers (i.e. radio access points above rooftop level) with the typical value of 500 m for ISD [12]. Finally, Ultra dense scenario corresponds to the 3GPP Dense Urban scenario that focuses on macro sites with or without micro sites and high user densities and traffic loads in city centers and dense urban areas. The key characteristics of this deployment scenario are high traffic loads (utilizing, both, mid band and mmWave frequency ranges, up to 1 GHz aggregated bandwidth), interference-limited, outdoor and outdoor-to-indoor coverage with the typical ISD value of 200 m [12].

The fronthaul segment in 5G/6G networks requires higher bandwidth compared to midhaul and backhaul due to its role in transmitting raw radio signals or IQ samples with minimal processing, resulting in substantial data rates and stringent low-latency requirements. This is in contrast to midhaul and backhaul, where data is aggregated and processed, leading to lower bandwidth demands [20]. ISD has a significant impact on the overall cost of fiber optic fronthaul. As the ISD increases, the probability of grouping multiple site locations on a common optical splitter (considering typical split ratios) decreases. This limits the potential for fiber deployment optimization, ultimately leading to higher deployment costs.

There is no one-solution-fits-all approach to meet all the previously stated x-haul requirements. Instead, each individual use case requires its own tailored solution. Besides this, ongoing research and emerging technologies introduce new capabilities and features that put pressure on x-haul networks in terms of additional requirements. The following paragraphs present emerging trends that should be taken into an account when considering x-haul requirements.

2.3. Cloud - RAN

C-RAN introduces the concept of cloud BBU, where various BBUs are centralized in different geographical areas. Consequently, the distances between RRH and BBU are very large, and this fact is an important driver in deciding which technology to utilize for the x-haul network. The centralized C-RAN concept is based on the migration of the processing functions of layers one, two and three to the central processor, namely BBU, and leaving the basic function of analogue/digital conversion to the RRH [4]. There are many possibilities how far this migration could go, depending on the desired level of processing functions at centralized units (CUs) and the requirements to distribute some of them at various distributed units (DUs). As the functional centralization level at CUs increases, bigger is the pressure on the fronthaul network in terms of capacity, latency, and reliability, even in case when user traffic is at minimum. Less centralized functional splits result with the more relaxed x-haul throughput and latency requirements as it depends on actual user traffic and user channel quality [4]. Conclusively, C-RAN and the corresponding fronthaul network should be designed jointly and optimally [4]. This means that the decision on which functional split to utilize highly depends on fronthaul/backhaul capabilities. When com-

pared to CoMP feature, functional split presents stricter when choosing x-haul solutions [4].

Besides the decision on functional splits, the required backhaul bandwidth, in a C-RAN architecture, depends on many factors such as aggregate carrier bandwidth, cell load, number of sectors, modulation and coding scheme, the number of antennae, and others [4].

2.4. Control/user plane split

Another important RAN disruption that had a significant consequence on x-haul network is Control/User Plane Split (CUPS). Small cells help to offload some of the capacities of the macro cells, thus providing better data rates to the users. On the other hand, small cells have significantly smaller coverage, resulting in more frequent handover processes, which is even more emphasized in ultra dense networks. The balance was found by splitting the user plane data, that is handled by the small cells, with the coordination and signaling data related to the control plane, that remains at the macro cell level and covers the selected group of small cells. This way, a handover process occurs only when the user leaves the coverage of the anchor macro cell. This architecture assumes that routers and switches are controlled logically by an external entity called software-defined network (SDN) controller or network operating system (NOS) [21]. CUPS, as well as C-RAN, requires very low latency in the backhaul segment to ensure coordination and synchronization, followed by effective backhaul throughput [4].

2.5. Softwarization

Among many things, network softwarization aspires to enable zero-touch service management, automation, and security. Software defined networking (SDN) and network function virtualization (NFV) are emerging trends that enable dynamic and programmable network configurations. Thanks to these features network can be easily automated and controlled, supporting multi radio access technologies (RAT) from multi-operators. SDN enables the control of not only the wavelength of the optical carrier, but also of the intermediate frequency over fiber at which the analog streams are transmitted, introducing enhanced granularity in subcarrier multiplexed transmissions [21].

X-haul is no exception, whether wireless or wired solutions are considered. Traditional optical access systems face inefficiencies due to long development cycles and high costs [22]. To improve this, a sharing architecture using common hardware and software is being developed. Projects like SDN-enabled broadband access (SEBA) and cloud central office (CloudCO) introduce virtualization and cloudification [22]. SEBA enables flexible service creation via software but currently supports only International Telecommunication Union (ITU-T) PON standards, requiring updates for IEEE PON standards compatibility [22].

The software-based approach reduces costs and simplifies management and it has been shown that these concepts can help to optimize backhaul resources and save up to 14 % of CAPEX [23,24]. SDN is based on the separation of user and control data that enables the introduction of heterogeneous x-haul links. It has proven to be a good solution to ensure optimized performance of the mobile x-haul network by dynamically finding the optimum x-haul route based on latency and available capacity, allocating required wavelength, and instantiating the location

of the local BBU [4]. Open software and white-box hardware facilitate scaling fiber access networks, transitioning from proprietary black-box solutions to open, interoperable systems [22]. Applications can request network resources dynamically through northbound interfaces, enabling multi-layer resource allocation across technologies [22]. SDN separates the control plane from the data plane, allowing centralized control and interoperability among multi-vendor network elements. It uses protocols like OpenFlow and NETCONF/YANG for communication between hardware and software. Incorporating optical layers into SDN minimizes latency and energy consumption, enabling dynamic, efficient bandwidth allocation [22]. Extending SDN concept to include optical parameters (e.g., wavelength) supports on-demand configurations for applications like mobile X-haul and cloud services.

The separation of control and data planes inherent to SDN fundamentally alters traditional transport network operations by introducing distinct management and forwarding domains. This decoupling increases the signaling load on the x-haul segments, as real-time network state updates and configuration changes must frequently traverse the physical infrastructure. Particularly for fronthaul transport, SDN-based architectures demand lower latency and support for agile bandwidth reconfiguration to ensure timely propagation of policies and dynamic adaptation to fluctuating traffic conditions as these requirements are critical for the orchestration of diverse mobile services [25,26].

Artificial intelligence (AI) and machine learning (ML) are increasingly used for fault management, resource allocation, and quality estimation in optical networks [22]. Digital twins bridge real-world and virtual environments for enhanced network optimization, while automation and distributed AI improve control plane efficiency in beyond-100G PONs, focusing on energy savings while maintaining quality of service through real-time feedback and control of PON infrastructure [22].

The main drawback of the SDN solution are security challenges due to control-user plane separation that exposes network to the prospective security breaches originating outside of the communication system [4]. SDN, as a concept, supports the convergence of the control plane of Optical Line Terminal (OLT) with the Radio Intelligent Controller (RIC). RIC is a critical component within the Open RAN (ORAN) architecture, enabling flexible and intelligent management of mobile network operations.

2.6. Energy efficiency

With the introduction of small cells, ultra-dense networks and all the additional features, the energy consumption of x-haul networks is no longer negligible. X-haul energy consumption has a significant share in the total energy budget, and it has been shown that fiber consumes less energy than microwave in a heterogeneous backhaul [4]. This is especially true in case of the PON utilization in fiber optic x-haul networks. As previously stated, a joint design of x-haul and access networks is proposed as the optimal solution for bandwidth allocation and latency. Energy consumption is no exception. The usage of heterogeneous radio elements and a cognitive radio x-haul approach enabled by self-organizing network (SON) capabilities proves to be an energy efficient solution [4]. The introduction of SDN controllers in the fronthaul infrastructure is able to provide an overview of the whole network, enabling the usage of simplified and low power consumption routers and switches [21].

Dynamic bandwidth allocation (DBA) optimizes resource distribution to Optical Network Units (ONUs) by adapting to real-time traffic demands and QoS requirements, while enabling power-saving modes during idle periods. This mechanism is critical in PONs, where adaptive DBA cycles and wavelength-sharing schemes minimize latency (under 100 μ s) and enhance bandwidth efficiency [14]. Radio-over-Fiber (RoF) integrates optical and wireless networks in fronthaul systems, balancing trade-offs between bandwidth, latency, complexity, and energy efficiency [14]. Advanced architectures like decomposed Arrayed Waveguide Grating Router (AWGR)-enabled Wavelength Division Multiplexing (WDM) fronthaul and hybrid Free Space Optics (FSO)/ millimeter wave

(mmWave) systems further optimize energy efficiency and scalability [14].

2.7. Network slicing

Network slicing has been introduced as a 5G feature that enables dynamic configuration of the independent virtual networks above the physical infrastructure. In this way it can tailor specific services according to the actual user's needs. This concept emphasizes the need for adaptive, configurable, and dynamic network based on appropriate QoS, low latency and sufficient bandwidth. X-haul needs to follow these requirements dynamically.

Traditional PON systems support multiple services through service isolation, Quality of Service (QoS) control, and connectivity configurations. According to the ITU-T G.9804.2 standard, the transmission convergence layer enables traffic management on the OLT through a single control function at the traffic level [22]. Slicing technology groups individual streams into slices on the OLT, adding control for sharing common resources and enabling service or virtual network isolation [22]. These slices are distributed across one or more ONUs and processed via the hierarchical scheduler in the DBA on the OLT [22]. In the future, parameters and slice profiles will need further study to optimize resources, particularly for applications such as network slicing with different delay limits [22].

The x-haul design should support network slicing requirements that are based on major parameters such as slice traffic, slice latency thresholds, network capacity, and underlying physical network's link bandwidth and length [27]. Additional parameters to consider when evaluating network slicing model are transport network traffic distribution as the slice traffic ratio distributed over three-layer networks (access, aggregation and core layer networks), substrate link load as the slice traffic to mapped physical link capacity ratio and, finally, serviceability as the ratio of the number of network slice instances (NSIs) which have been successfully mapped to the total number of instances, thus evaluating the ability of a network to serve simultaneously network slicing requests [27].

Research shows that network slicing in transport networks, while frequently emphasizing latency as a crucial service-level agreement (SLA) parameter, also imposes significant requirements on bandwidth, resource isolation, and dynamic traffic management [27]. Each slice may demand dedicated or parallel logical x-haul paths to guarantee SLA-defined characteristics such as bandwidth and isolation, which in turn increases both aggregate and peak demands on x-haul infrastructure and introduces pronounced burstiness in traffic patterns. These findings underline that the implementation of diverse slice-specific SLAs affects not just latency, but also shapes the overall network resource consumption and operational dynamics of multi-slice transport environments [27].

The number (or groups) of network slices and x-haul network latency are mutually proportional. Besides the number of network slices, it is important to consider network slice type as well. Authors in [27] presented several examples of the network slice types and their impact on x-haul.

Small-volume, critical communications (s-VCC), from the family of mMTC service types, is typical for services such as robotic control and industry control and they require high reliability, millisecond latency and small traffic volume. Most of the traffic for this network slice type occurs at the access layer, therefore the biggest pressure in terms of network slicing requirements would be on the fronthaul [27].

Extreme real-time communications (eRTC) belong to the URLLC family of service types that enable services such as autonomous driving, driving assistant or automotive factory. These services require latency below one millisecond, mobility, and high traffic volume. Aggregation layer network must bear the biggest pressure in terms of resource allocation (approximately 55 % of the traffic ratio) for this network slice type, while access and core layer equally handle rest of the traffic [27].

Enhanced mobile broadband access in dense areas (eMBB) offers services such as hologram, high-definition (HD) video, user mobile broadband in a stadium and it is characterized by high traffic volume and high throughput. In this case aggregation and core network layer are equally loaded, putting the biggest pressure on midhaul and backhaul networks [27].

Physical resource utilization in the access layer is practically constant and it is irrelevant to the percentage of node connectivity per network, while at the aggregation and core layers it decreases as the percentage of node connectivity per network increases [27]. The increase of network serviceability is followed by the increased network latency's threshold [27].

2.8. Additional 6G requirements

6G mobile networks have a need for additional requirements on top of all the previously mentioned ones that were defined for 5G mobile networks. As 6G is still in early phase of its evolutionary path it is difficult to exactly list all the requirements. Nevertheless, in this section, the most obvious ones that have direct influence on x-haul network will be mentioned.

Since lower and mid bands are overloaded with user traffic, it is conceivable to assume that 6G will continue to move to higher frequencies with wider system bandwidth. The Frequency Range 3 (FR3) 7 - 24 GHz is key for 6G networks as it provides an optimal balance between propagation and available spectrum, enabling a significant capacity increase [28]. Its lower range (7 - 8 GHz) supports acceptable indoor penetration and coverage similar to today's C-band, allowing cost-effective reuse of existing sites. FR3 offers wide contiguous bandwidth, supports densely packed MIMO antennas for high spatial multiplexing gains, and enables gigabit-per-second throughput over kilometer-scale distances with lower power consumption and reduced hardware complexity compared to higher frequencies [28]. These features make FR3 the main workhorse for scalable, high-capacity, and cost-effective 6G connectivity [28]. mmWave and THz frequency ranges from 100 GHz to 3 THz are also promising bands for the next generation of wireless communication systems [29]. The THz band (300 GHz to 3 THz) is particularly interesting as it is characterized with ultra-wideband (UWB) resources and it can support ultrahigh wireless communication rates ranging from 100 Gb/s to even 1 Tb/s, with the increased peak rate of existing 5G by one or two orders of magnitude [30]. With its UWB resources the THz band enables emerging applications, such as ultrahigh-resolution holographic communication and metaverse with capacity exceeding 100 Gb/s [30]. This data rate is close to the fiber channel capacity, therefore the challenges related to the processing of UWB wireless signals could be expected, because of the limitations of the bandwidth, sampling rate, and accuracy of highspeed digital-to-analog and analog-to-digital converters (DAC/ADC) [30]. This brings us to the general conclusion that THz communications will be seriously challenged by inherent limitations of solid-state electronic devices in terms of bandwidth, characteristic frequency, and power efficiency [30].

The coverage of the THz cells will be significantly limited due to its large air propagation loss, weak diffraction, and diffraction capability, resulting with the requirement for denser THz infrastructure deployment [30]. These short wavelengths at mmWave and THz bands will allow massive spatial multiplexing in hub and backhaul communications [29]. Ultra dense heterogeneous 6G networks will support low-frequency, microwave, mmWave, and THz bands, which require the deployment of ultra-high-capacity x-haul infrastructures with the optical fiber as the most suitable solution, although THz wireless backhaul will be an attractive alternative as well [29].

Further 6G research proposes the introduction of distributed arrays in THz channels to increase the achievable rates and coverage, that is, to introduce reconfigurable intelligent surfaces (RISs) [31]. It is reasonable to think of the transport links of unspecified bandwidth between base stations and each RIS as a new segment of the backhaul network that is supposed to play a key role in successful operation of RISs [31].

Transport bandwidth is defined according to the number of elements at the RISs, the number of control bits per element, and their refresh rates [31]. Consequently, this will increase the control plane latency of the overall system [31]. The transport network capacity will scale in proportion with the number of the served user equipment, while the required number of RISs per each cell remains unclear [31].

When it comes to system latency, it is mainly determined by, among other things, the x-haul network and edge-cloud capabilities [31]. Edge-cloud close to the radio access network can help to reduce the required length of the x-haul network, thus enabling lower latency of the overall system.

6G networks will also make another step further from network softwarization to network intelligentization as it will be required to support artificial intelligence (AI) applications from the core to the end devices of the network [29]. This means that network will have to support diverse capabilities, including communications, content caching, computing, and even wireless power transfer [3]. The aim of the intelligentization is to make network to be smart, agile, and able to learn and adapt itself according to the changing network dynamics, namely, to evolve into a "network of subnetworks," allowing more efficient and flexible upgrades, as also to support new framework based on intelligent radio and algorithm-hardware separation [3]. Consequently, the nature of mobile terminals is supposed to change, with mobile robots and vehicles playing a more important role, therefore stronger dependencies between networking infrastructures and applications are expected [29]. AI enables advanced optimization that includes latency, bandwidth, and energy efficiency, making the network management intelligent, enabling dynamic resource allocation, predictive maintenance, and fault detection [14]. Machine Learning (ML) driven traffic prediction models optimize 6G fronthaul networks by dynamically adjusting interface splitting options based on real-time demand, maximizing resource efficiency [14]. AI/ML enhances performance by analyzing vast datasets to predict network behavior, enabling precise resource allocation and adaptability to evolving data patterns [14]. These systems automate network adjustments, minimizing manual intervention while maintaining responsiveness to 6G's dynamic requirements. This integration of AI/ML drives smarter, self-optimizing networks tailored for next-generation connectivity [14]. However, challenges remain, above all in terms of quality and quantity of data, complexity of ML models and cost of deployment [14].

mMIMO will continue to be the technology of the vital importance for 6G to increase spectral efficiency [29]. Furthermore, with the increased data rates and reduced delays it is expected that a significant amount of the computational and storage functions will be moved to the cloud [29]. This will have a direct consequence on the capacity planning of the links between mobile access network and corresponding central offices or data centers, namely fronthaul, midhaul and particularly backhaul segments.

Finally, underwater mobile communications are envisaged as a novel use case of the 6G networks. The intention is to provide coverage and Internet services for broad-sea and deep-sea activities, mostly for military or commercial applications [29]. Consequently, it can be expected that backhaul segment of the architecture will be extended into the underwater fiber optic cables and networks.

All these additional requirements are directly influencing network parameters that are involved in the x-haul link capacity planning such as desired bandwidth, applied MIMO scheme, the number of sectors of the observed mobile network infrastructure point or maximum mobile peak bit rate.

3. Review of current solutions and emerging trends in x-haul networks

X-haul networks can be divided in two main groups: wired and wireless x-haul networks. Within these two groups there are many possible solutions.

3.1. Wired X-haul solutions

Fiber optic networks represent the most prevalent example of wired x-haul solutions and are widely recognized as the optimal choice from a technological standpoint. Their advantages include immense bandwidth, exceptional reliability, low latency, high capacity, extensive coverage, superior bit error rate performance, and immunity to environmental interference. Despite these benefits, the high costs associated with deploying fiber optic networks remain a significant barrier. Installation processes are both time-consuming and expensive, and in some cases, deploying fiber optics in remote locations, such as mountainous regions or isolated islands, is not feasible. In this paper, the term “fiber optic network” refers to two primary deployment models. The first model involves point-to-point networks, typically utilizing Fiber Ethernet technology, while the second comprises PON based on a point-to-multipoint topology.

An alternative wired x-haul option involves the use of copper networks, which support access technologies such as Ethernet and Digital Subscriber Line (DSL). However, compared to fiber optic networks, copper-based solutions offer significantly reduced range and capacity and are highly prone to environmental interference. As a result, copper networks are increasingly considered outdated, as they are no longer capable of supporting the advanced features of modern 4G and 5G networks.

The discussion in Section 4 focuses specifically on wired x-haul solutions, with an emphasis on fiber optic technologies.

3.2. Wireless X-haul solutions

There are more options available in wireless x-haul solutions when compared to wired ones. Wireless x-haul is a cost-effective solution, but its deployment highly depends on various parameters such as traffic intensity, propagation conditions, cost factor, site locations, and interference conditions [15]. Microwave and millimeter wave (mmWave) as wireless backhaul medium are the main representatives of this solution.

mmWave covers extremely high frequency ranges 30–300 GHz, mainly used for small cell backhaul solutions, while lower bands, typically in the range 6 GHz to 60 GHz band, are also utilized for radio frequency (RF) mmWave backhaul networks [15]. Since mmWave-based backhaul utilizes high frequency ranges it can offer high data rates (1–2 Gbps), but at the expense of shorter reach when compared to microwave backhaul solution. Small reach is due to high propagation losses, which are mostly a consequence of multipath propagation and fading due to rain absorption. Since low wavelengths are utilized, the absorption of the mmWave signals by water molecules are a significant factor to consider during the deployment in more humid areas. These challenges can be relaxed to a certain extent thanks to the introduction of MIMO technique. The propagation is based on narrow beams therefore, the emphasis should be put on precise and stable alignment. Due to the limited coverage of mmWave connectivity, line of sight is required for a backhaul/fronthaul fixed link between two endpoints. Another emerging trend is in-bound backhauling that assumes reusing the hardware and spectrum from the radio access for wireless backhaul links, thus, increasing resource efficiency and reducing capital expenditures (CAPEX) [4].

C-RAN architecture requirements are practically satisfied by deploying fiber optic solution in the x-haul segment. However, existing fiber optic infrastructure is a very scarce resource and significant CAPEX is required to build suitable fiber optic x-haul network. mmWave x-haul can be regarded as a suitable alternative, although propagation challenges significantly affect system's resilience.

Microwave-based wireless x-haul links enable maximum reach of approximately 50 km at up to 10 Gbps data rates [15]. The parameters related to MIMO schemes, advanced modulation with 4096 quadrature amplitude modulation (QAM) and wider channel spacing (112 MHz at 4–42 GHz bands) support high spectral efficiency in microwave links,

thus enabling higher data rates. Alongside with the fiber optic, microwave is a widely used solution for mobile x-haul traffic [15]. It is less expensive, and it takes lower deployment time, especially in cases when it covers more small cells. However, that could increase the utilization of the shared bandwidth and may result in throughput degradation [4]. On the other hand, it is highly susceptible to weather conditions and propagation environment.

In the context of the 6G networks, THz wireless x-haul systems can support bandwidth, latency, and reach requirements. THz wireless systems operate in the frequency range between 300 GHz and 3 THz, offering huge bandwidths of several tens of GHz. The first standard for THz communication (IEEE 802.15.3d) considers transmission rate up to 100 Gb/s, while further research focused on THz technology aim at MIMO systems for THz communications enabling up to 1000 Tb/s transmissions [21]. This solution faces two major challenges. The first one is the limited reach. Obstacles like rain, humidity, trees, poles, and concrete walls can block THz signals due to their very short wavelengths. It has been shown that THz systems can support up to 100 m links with directional antennas and up to 10 m with omnidirectional antenna [21]. Another challenge arises from the inherent limitations of solid-state electronic devices. Recent research in semiconductor technology has made communication systems beyond 275 GHz possible [21,30].

Among many alternatives for wireless x-haul are free space optics (FSO). FSO is a line-of-sight technology that utilizes a light beam generated at semiconductor lasers for data transmission. Consequently, it is resilient to electromagnetic interference that can be generated by various sources from environment. It is like the fiber optic solution as FSO systems also transmit light, but instead through a fiber, it utilizes the air or free space as medium. The main advantages of this solution are low power consumption, high capacity, low price, the usage of license-free spectrum, and it can be used to set up point-to-multipoint links [1]. FSO requires Line-of-sight (LoS) conditions and it operates in the unlicensed 800–1700 (nm) wavelength range (above 300 GHz) [14]. Due to the densification requirements for 5G/6G networks and reduced distances among sites to a few hundred meters, it proves to be a promising solution as a x-haul technology. FSO can support splitting options ranging from 1 to 6 and 7.3 [14]. When compared to fiber-based fronthaul, it can offer a 50 % reduction of deployment costs, or even higher for areas with very low or non-existent level of available fiber infrastructure [1]. It has been shown that FSO solutions can achieve up to hundreds of Gb/s data rates, going up to Tb/s via a wavelength sharing technique [21]. It is also compatible with legacy backhaul technologies and it can be deployed as a part of hybrid solutions such as mmWave/FSO [21]. The negative side of the FSO solution is the susceptibility to adverse weather conditions such as fog, snowfall, or rain, making it unreliable solution for long distances, especially for non-line of sight (NLoS) links [1]. Consequently, FSO usage is limited by absorption and dispersion, scattering, atmospheric attenuation, scintillation and geometric loss [14]. Recent research focused on these challenges has proposed some mitigation solutions based on diversity and the AI or machine learning (ML) applications [21].

Another novel approach are satellite networks in mobile x-haul. The convergence of satellite networks with terrestrial wireless networks allows seamless connectivity and improved service continuity, particularly in areas where terrestrial infrastructure is sparse or during network outages. Advances in satellite technologies, such as High Throughput Satellites (HTS) and Low Earth Orbit (LEO) made satellite backhaul a competitive option.

4. Fiber optics as a x-haul solution in 5G/6G networks

This section revisits the analysis initially presented in the FOAN workshop paper [5], which focused on fiber optic x-haul solutions. Here, the discussion is expanded by including additional technical details and addressing broader x-haul solutions.

As mentioned earlier, fiber optic technology remains the most advantageous choice for x-haul networks, as it meets the majority of the requirements outlined in Section 2. However, a significant challenge associated with fiber optics is its high cost and limited availability. Millimeter wave (mmWave) and free-space optics (FSO) represent the only viable wireless alternatives, while traditional microwave backhaul is increasingly seen as inadequate in light of the growing traffic demands of 4G and 5G networks [4].

4.1. Topology

Fiber optic x-haul topologies can use Point-to-Point (PtP) or Point-to-Multi-Point (PtMP) configurations. PtP links can be deployed in chain, tree, ring, or mesh networks, but one has to bear in mind that link length and latency are proportional to each other. It offers higher security and easier OAM but is more expensive, making it essential for fronthaul in C-RAN splitting options that support greater centralization levels [1]. Novel PtP standards assume the transmission of optical access signals bidirectionally (BiDi) via a single fiber while other PtP standards (e.g. Ethernet, SONET, etc.) often assume the use of two fibers to connect the link, each for one direction in transmission [32]. ITU-T has defined in its Recommendation G.9806 Amendment 3 BiDi PtP link specifications supporting 10, 25, and 50 Gb/s data rates, while specifications for 100 Gb/s BiDi PtP links is under consideration [11]. These specifications enable total reach from 10 km up to 40 km [11].

PtMP is cost-effective but can introduce higher latency and lower security due to shared medium use. PtMP can be regarded as preferred solution for mobile fronthaul in C-RAN architectures [4]. It has been shown that PON solution that is based on PtMP topology reduces up to 60 % of 5G mobile backhaul cost [33]. Key considerations for optimal PtMP use include RRH group formation connected to a single PON splitter, optimal BBU pool placement to minimize links costs, and path optimization from the RRH to its corresponding splitter and from the splitter to the corresponding BBU pool [1].

4.2. Enabling fiber optic technologies

There are several technology solutions that can be implemented in C-RAN networks, which could drive the evolution of fronthaul towards 5G and beyond, including space-division multiplexing (SDM), multi-core fiber (MCF), and MIMO signaling [21].

SDM technology in optical MCFs addresses the growing demands for fronthaul network capacity, peak data rates, and flexibility [21]. SDM-based C-RAN networks enable higher data rates and parallel connections between the central office (CO) and RRHs, supporting reconfigurable MIMO transmissions and remote beamforming [21]. They also allow both digital radio-over-fiber (D-RoF) and analog radio-over-fiber (A-RoF) signal transmissions within the same network, and interoperability with PONs [21]. A-RoF fronthaul generates radio frequency signals at the CO, providing larger modulation bandwidths in the optical domain, leading to a higher level of centralization of critical equipment like network control and radio frequency oscillators [21].

Ribbon optical fibers, offering a reduced physical footprint due to higher fiber density, can serve as an alternative to MCFs in the fronthaul segment [21]. However, MCF remains the preferred solution for advanced optical and radio technologies, such as optical beamforming and joint signal control and amplification, especially in SDM and wavelength division multiplexing (WDM) transmission [21].

Ultra-dense WDM technology, combined with reconfigurable optical add-drop multiplexers (ROADMs) based on photonic integrated circuits, provides a promising solution for fiber fronthaul. It offers intelligent, reconfigurable features while remaining compatible with existing x-haul infrastructure [21].

4.3. PON solutions

Advancements in PON standards focus on improving medium sharing and spectral efficiency. When PON transport and detection mechanisms are concerned, coherent PONs offer advantages over intensity modulation and direct detection (IM-DD) systems in terms of extended reach and improved GPON coexistence. However, these benefits come at the cost of higher complexity, increased power consumption, and greater expense [34]. All IM-DD solutions must adhere to the 29 dB PON optical power/loss budget to ensure compatibility with the existing fiber infrastructure, presenting a significant challenge [22].

PON networks meet the mobile access network rate demands thanks to the supported modulations. For very fast IM-DD Next-Generation Passive Optical Network Stage 2 (NG-PON2) ITU standard (beyond 100G), PAM4/M modulation is a promising solution due to its low cost and complexity compared to non-return-to-zero (NRZ) modulation [22]. Advanced modulation formats include multi-carrier orthogonal frequency division multiplexing (OFDM), single-carrier quadrature phase shift keying (QPSK), and high-order quadrature amplitude modulation (QAM) modulation [22]. To support higher data rates and to improve the spectral efficiency, coherent transmission is introduced supporting advanced modulation formats that can be encoded in all four dimensions of the light field: in-phase and orthogonal in the two orthogonal polarization states [22]. NG-PON2 systems are limited by chromatic dispersion, which can be mitigated by using the O-band for transmission. To improve sensitivity and optical budget, semiconductor optical amplifier (SOA) and erbium-doped optical fiber amplifier (EDFA) amplifiers can be used, with SOA being particularly suitable due to its operation in the C and O bands, easy integration, and low cost [22].

Another goal is to increase the splitting ratio without compromising mobile network needs, with the ratio expanding from 1:64 (G.984.1 standard) to 1:128. Alternatively, direct fibers from the feeder cable, not terminating at passive splitters, can connect large macro base stations, but this is less cost-efficient solution.

The main step towards emerging optical infrastructures has been to make fixed grid a flexible [35]. Fixed grid is characterized by equal channel spacing and spectrum allocation, regardless of its capacity and actual demand, which results in the loss of significant parts of the spectrum [35]. Once a network is adaptable to traffic or network conditions (such as spacing between channels or types of modulation), then it can be considered as a flexible network that supports more efficient spectrum utilization and growing traffic demands [35].

The following paragraphs discuss PON solutions in the context of 5G/6G x-haul, following their evolution as shown in Fig. 2. It assumes that compliance with the NG-PON2 standard ensures the fulfillment of 5G/6G RAN requirements [8]. Each PON solution is evaluated based on its alignment with NG-PON2, with a summary of integrated PON solutions in the x-haul network provided in Fig. 3.

Time Division Multiplexing - Passive Optical Network (TDM-PON) systems, based on Gigabit Passive Optical Network (GPON) or Gigabit Ethernet Passive Optical Network (G-EPON) technologies, operate within PtMP topology and offer data rates of 1 Gbit/s, with potential upgrades to 10 Gbit/s through XG(S)-PON and 10G-EPON technologies [6,7]. Surpassing the nominal line rate of 10 Gb/s, the IEEE 802.3ca 50G Ethernet Passive Optical Network (EPON) Task Force has recently established a system with a nominal line rate of 25 Gb/s, while the ITU-T has advanced further by adopting a nominal line rate of 50 Gb/s [36]. The NG-PON2 ITU standard introduces upgrades for TDM-PON, allowing data rates to increase up to 40 Gbit/s [10]. This has been followed by ITU-T Higher Speed Passive Optical Network (HS-PON) system project, supporting up to 50 Gb/s line rate per wavelength channel [36]. 50G-PON utilizes TDM/TDMA and supports a 50 Gb/s downstream line rate and two upstream line rate options (25 Gb/s and 12.5 Gb/s). It is a robust solution that comes with high efficiency while it enables co-existence with existing G-PON or XG(S)-PON solutions for backward compatibility purposes [36]. Recently, the Full-Service

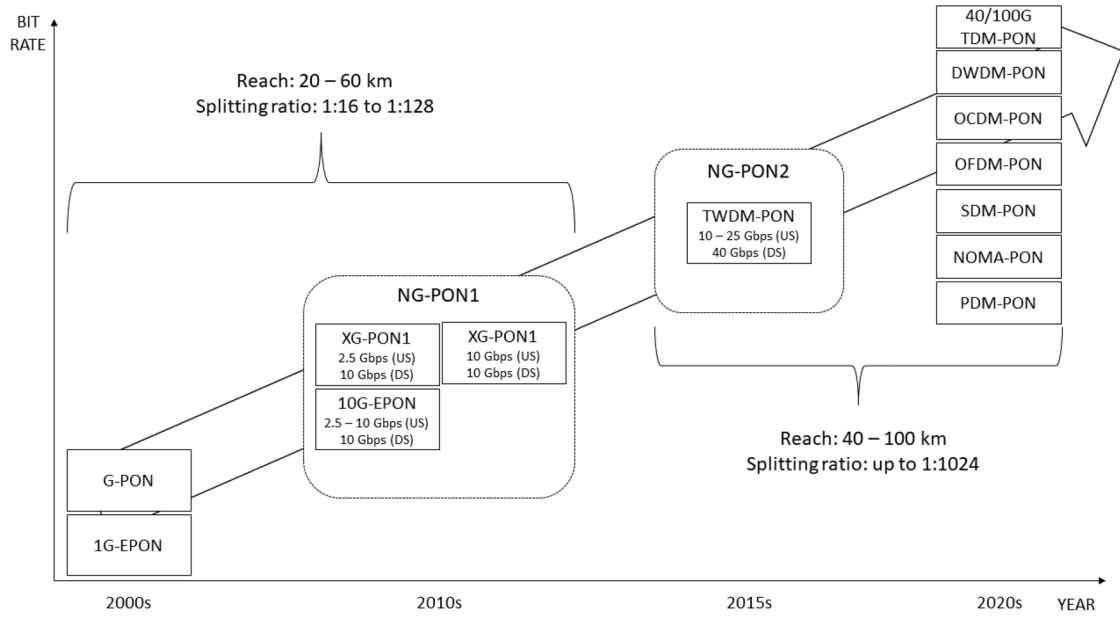


Fig. 2. Evolution of PON and its standards.

Access Network (FSAN) community and ITU-T SG15 study group have focused on PON solutions capable of realizing line rates beyond 50 Gbps within the very high-speed PON (VHSP) project. VHSP project assumes PON solutions providing not just residential services, but also broadband services to enterprises and antenna sites supporting line rates from 100 Gbps up to 200 Gbps, followed by 20 km up to 40 km reach [34]. The expected deployment of the 200 Gbps solutions is expected to be in roughly 2035 [32]. For 100G/200G-PONs IM-DD is still the preferred solution, but nevertheless, coherent technology must be seriously considered for 40 km reach and above since the limits of IM-DD for TDM PON are approached [34]. Alongside with the increased data rate of IM-DD PONs, the receiver sensitivity also degrades from -27 dBm at 50 Gb/s to -20 dBm and -16 dBm at 100 Gb/s and 200 Gb/s, respectively [32]. Coherent PONs, on the other hand, are able to support 200 Gb/s per wavelength with a receiver sensitivity of -32 dBm [32]. While TDM-PON is able to meet the capacity demands of 5G/6G fronthaul, latency, particularly in upstream directions, remains a challenge due to shared bandwidth and time-slot access [4,14]. Dynamic bandwidth allocation (DBA) algorithms can reduce latency, but issues like power budget limitations, component availability and maturity, high costs associated with high-speed transmitters and receivers, and chromatic dispersion still hinder performance [37].

Wavelength Division Multiplexing - Passive Optical Network (WDM-PON) systems are deployed as PtMP topology with dedicated wavelengths for each Optical Network Unit (ONU) [9]. Compared to TDM-PON, WDM-PON offers higher scalability, reduced latency, and is more dynamic and adaptive, but is more costly due to the need for stable and precise components like expensive transceivers and multiplexers, which rely on expensive laser devices and wavelength-selective switches. It meets 5G fronthaul bandwidth requirements, as each wavelength is able to offer the capacity of at least 25 Gbps [14]. ITU-T G.9802.2 standard defines WDM-PON system with 20 wavelength channels in the C-band, divided into two spectral ranges, namely downstream at 1529.553–1544.526 nm and upstream at 1550.116–1565.496 nm with channel spacing of 100 GHz (uniform in both directions) [32]. It specifies loss budget classes X of 13 dB and Class Y of 17 dB, that support 10 km and 20 km reach for wireless fronthaul transport, respectively [32]. Variants like Dense WDM (DWDM) and Ultra-Dense WDM (UWDM) offer higher bandwidths that are able to meet 5G/6G x-haul requirements, but at a higher cost, while Coarse WDM (CWDM) is a more

affordable option. Despite these benefits, WDM-PON faces challenges in meeting NG-PON2 requirements, particularly regarding backward compatibility and therefore it is less viable for 5G/6G x-haul [37].

Time and Wavelength Division Multiplexing - Passive Optical Network (TWDM-PON) is a hybrid system combining TDM-PON and WDM-PON architectures, offering up to 40 Gbit/s throughput based on the NG-PON2 standard [8]. In this system, fronthaul aggregation points are managed as WDM-PON, while ONUs connected to each fronthaul aggregation node operate under a TDM-PON architecture. It balances between the simplicity and cost-efficiency of the TDM-PON and flexibility of the WDM-PON [4]. A typical use case for TWDM-PON is connecting the DU and RU to meet the bandwidth requirements for a mMIMO fronthaul network at a reduced cost [38]. Each ONU is equipped with tunable transmitters and receivers, allowing multiple ONUs to share the same wavelength but in different time slots, with each ONU selecting a wavelength for a specific time slot [38]. With the application of several DBA schemes and with the efficient use of wavelengths, antennas, and resource blocks it is able to improve bandwidth distribution and to minimize fronthaul load and energy consumption [38]. TWDM-PON is a highly adaptable technology where each channel can operate at different data rates, efficiently addressing varying bandwidth requirements. This flexibility allows seamless support for both legacy and new technologies, making it an ideal solution for the stringent fronthaul demands of 5G networks [14]. The FSAN community has selected TWDM-PON as a highly promising candidate for 5G/6G x-haul [37].

One promising solution for achieving throughput of up to 100 Gbit/s is the use of Orthogonal Frequency Division Multiplexing - Passive Optical Network (OFDM-PON) for downstream traffic, combined with TDM-PON for upstream traffic. OFDM-PON is capable to offer software-defined upgrades in terms of bandwidth flexibility, system scalability, and achievable data rates on a per-wavelength basis thanks to its underlying digital signal processing (DSP)-based implementation [39]. It is not a suitable solution for upstream since transmissions are uncoordinated, but in downstream it creates low latency, virtual PtP links over individual wavelengths to serve high density of cells with sufficient bandwidth [39]. It is a very attractive solution from NG-PON2 point of view because of its strong dispersion tolerance and high spectral efficiency, however, due to its high DSP complexity it is still not viable for full commercial use [22]. OFDM-PON offers a significant advantage in enhancing bandwidth utilization over long distances compared to other

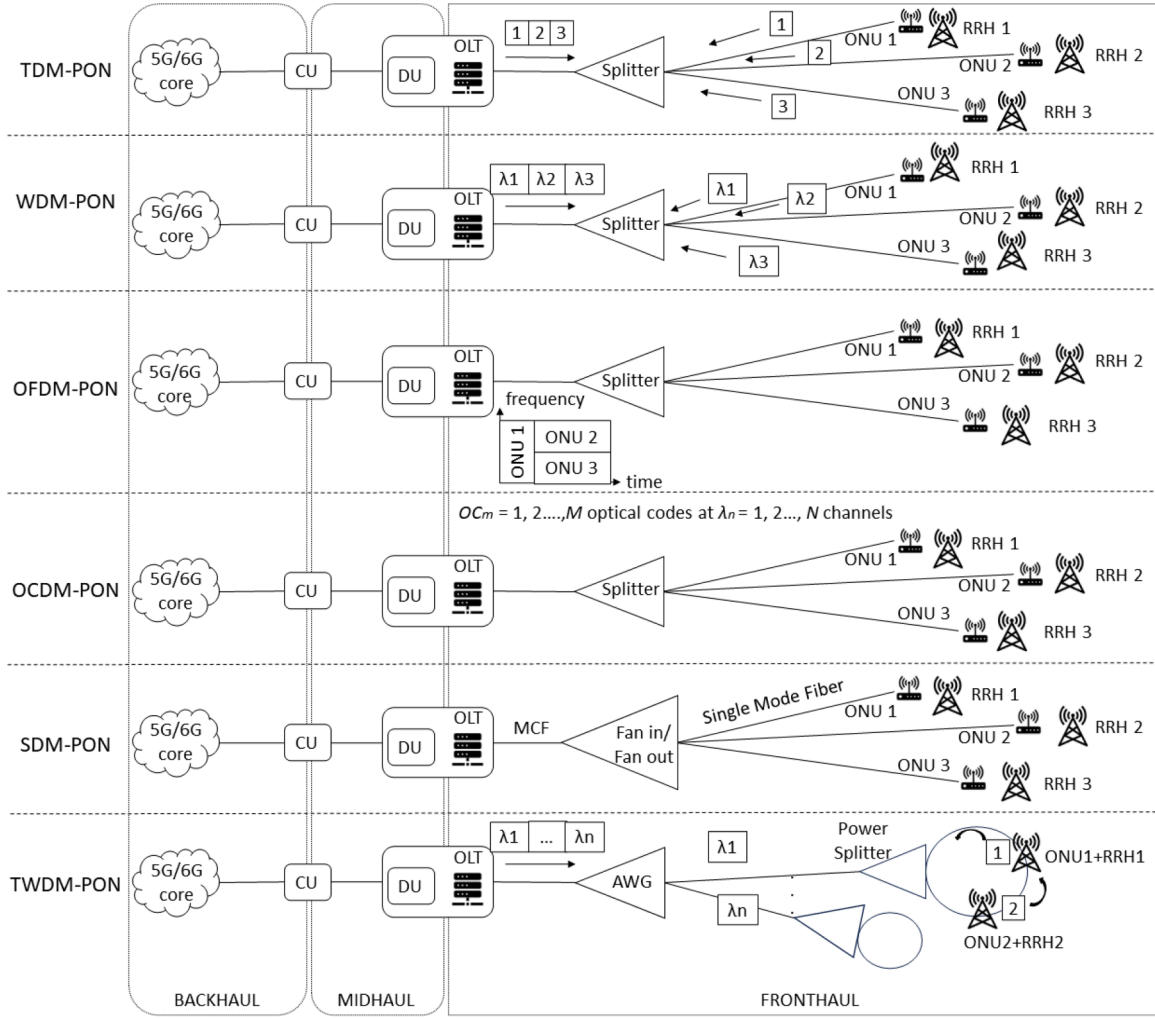


Fig. 3. A summary of the observed PON solutions embedded in the x-haul network.

PON technologies like TWDM-PON which makes it an ideal solution for advanced applications such as 5G and 6G fronthaul, particularly when ultra-high capacities of at least Tbps are required [14]. Its high cost and limited availability of technical components prevent it from meeting NG-PON2 requirements, though it remains a promising solution for 5G/6G x-haul as it matures [37].

The Optical-Code-Division Multiple Access (OCDMA) technique in PON networks performs encoding and decoding in a PON through optical codes thus allows the selection of the desired signal so that different ONU users can share the same bandwidth [22]. It offers additional advantages such as efficient bandwidth utilization, interference resilience, enhanced security, network flexibility and support for asynchronous operation. Unlike TDM-PON, OCDMA provides user access without the need for traffic management or system synchronization and it also eliminates the need for complex hardware like wavelength-controlled tunable lasers, which are required in WDM-PON systems. Although OCDMA-PON is considered a promising solution for x-haul networks, it is currently impractical due to its inability to meet all NG-PON2 requirements, mainly due to the immaturity of key components such as photonic encoders and decoders [37].

Space Division Multiplexing (SDM) has been recognized for its ability to increase the capacity of optical networks, particularly in long-haul applications. SDM enhances optical fiber capacity by dividing its cross-sectional area into multiple spatial channels, each carrying a distinct data stream. These channels are created using techniques such as mul-

ticore fibers (MCFs), which contain multiple cores, or few-mode fibers (FMFs), which utilize multiple modes within a single core [14]. By leveraging these spatial channels, SDM significantly increases data transmission capacity compared to traditional single-channel optical fibers [14]. It has been shown that Space Division Multiplexing - Passive Optical Networks (SDM-PON) can be a cost-effective 5G/6G fronthaul solution [40]. In the paper [40] the authors have demonstrated the successful operation of 100G Ethernet over a MCF in a PtP topology. Typically, SDM requires complex MIMO digital signal processing (DSP), which limits its practical use in the near future. However, the proposed SDM-PON architecture uses weakly coupled MCF, where each core functions as a parallel P2P Ethernet link, eliminating the need for complex DSP. This allows for data rates of 10 Gb/s or higher per user which satisfies NG-PON2 requirements [40]. In conclusion, although SDM-PON offers significant advantages in terms of capacity, energy efficiency, and scalability, further research, development, and cost reduction of the technology, as well as integration with existing network infrastructure standards, will be necessary for its widespread application in 5G/6G x-haul networks.

Non-Orthogonal Multiple Access - Passive Optical Networks (NOMA-PON) integrates Non-Orthogonal Multiple Access (NOMA) with OFDM to enhance the performance of PONs. NOMA-PON allows multiple ONU users to share the same frequency resources simultaneously by transmitting signals with different power levels, resulting in overlapping signals in the time and frequency domains and multiplexing in the power

domain [22]. NOMA-PON is a new approach, integrating NOMA with OFDM to enhance optical network performance [14]. OFDM divides bandwidth into orthogonal subcarriers, enabling high data rates while mitigating chromatic dispersion and intersymbol interference. Unlike conventional OMA-based systems, which allocate distinct subcarriers to different ONUs, NOMA-PON employs power-domain NOMA [14]. This allows multiple ONUs to share the same subcarrier via varying power levels, significantly improving user capacity and data throughput [14]. Successive interference cancellation (SIC), based on multi-user detection technology, effectively distinguishes and separates user signals at the receiver based on power differences. This approach enhances data throughput and spectral efficiency without requiring additional resources, enabling more users to be served within the same time and frequency range [22]. By assigning different power levels to multiple ONUs on the same subcarrier, it improves spectral efficiency, user fairness, and data rate capacity compared to traditional Orthogonal Multiple Access (OMA) approaches [14]. NOMA-PON has demonstrated superior performance over conventional OFDM-PONs, particularly in terms of power budget and data rate, making it a promising solution for future 6G fronthaul networks [14]. However, challenges remain, including mitigating Self-Induced Intermodulation Interference (SSII) caused by overlapping signals with varying power levels and developing effective multiple access control protocols to coordinate ONU transmissions efficiently [14]. With its potential to deliver high spectral efficiency, scalability, and robust network architectures, NOMA-PON is well-suited for NG-PON2 networks. However, further research and optimization are required to address its current limitations and realize its full potential in practical deployments, i.e. 5G/6G x-haul [14].

Polarization Division Multiplexing - Passive Optical Network (PDM-PON) utilizes polarization states to increase the data rate and spectral efficiency of optical signals, achieving capacities of 100 Gb/s or more per wavelength [14]. By modulating signals with distinct polarization states and employing coherent detection at the receiver, PDM-PON supports high-capacity optical access networks. PDM-PAM-4 is a modulation format that utilizes four-level pulse amplitude modulation, encoding two bits per symbol [14]. It offers high receiver sensitivity, colorless frequency selectivity, and linear detection, enabling compensation for channel impairments in the digital domain [14]. To reduce system costs and power consumption, PDM-PON can leverage low-cost, low-complexity components, including intensity modulators, distributed feedback (DFB) lasers, and Kramers-Kronig receivers [14]. To fully meet the stringent demands of 5G/6G fronthaul networks such as high capacity, low latency, and flexible resource allocation further innovations in advanced modulation formats, coherent detection methods, and joint resource allocation schemes are necessary [14]. Despite these challenges, PDM-PON represents a promising approach for high-performance optical access networks, therefore it can be considered as a potential solution for 5G/6G x-haul [14].

5. Conclusions

This paper presents the main x-haul requirements arising from the 5G/6G radio access network standards. It also provides an overview of the available solutions that satisfy these requirements. Each RAN deployment comes with different requirements, consequently, a specifically tailored x-haul solutions should be considered. In addition to mobile traffic, coverage, latency and energy consumption as standard set of requirements, emerging technology trends that come with 5G/6G standards such as C-RAN, CUPS, softwareization, network slicing, THz bands, RIS and intelligentization are also considered.

An overview of current and emerging solutions in x-haul networks is given. From the point of view of wireless x-haul, microwave, mmWave, THz and FSO x-haul are highlighted as promising solutions, but the focus is on fiber optic x-haul. Fibre optic x-haul is a future-proof solution that prevails the best for all requirements, the only exception being cost. The evolution of the fiber optic fronthaul in the context of C-RAN networks

is also based on many available technologies like SDM, MCF, ribbon optical fibers, ultra-dense WDM and ROADM.

PON solutions stand out because they can be considered the most cost-effective fiber optic solution. It is assumed that the compliance of the x-haul network with the NG-PON2 standard ensures the fulfillment of all requirements arising from the 5G/6G RAN segment. All available PON solutions are listed chronologically, and each one is viewed against the requirements of NG-PON2 as far as possible. It ranges from TDM-PON up to the OCDM-PON and OFDM-PON, but in the context of next-generation radio access networks, TWDM-PON is expected to be the most promising solution for mobile fronthaul due to its large capacity and flexibility between time and wavelength resources.

CRedit authorship contribution statement

Damir Breskovic: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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